

Problem F: Optimal Mirroring

Description

Nanami Touko is obsessed with the concept of a “perfect performance.” To her, identity is like a permutation $a_1, a_2, \dots, a_n$ of $1, 2, \dots, n$. She believes her late sister was the original, perfect permutation a . Now, Touko spends her days trying to act in a way that is “similar” to that ideal, ensuring that in every scene, her relative “shape” matches her sister’s perfectly.

Koito Yuu, the only one who truly sees the girl behind the performance, wants to help Touko organize her various scripts. Touko has a collection of q scripts, represented as intervals $[l_i, r_i]$ with $1 \leq l_i \leq r_i \leq n$.

To analyze these performances, Yuu uses the following definitions:

- A performance $b_1, b_2, \dots, b_n$ **holds its shape** for an interval $[l', r']$ if for any pair of integers (x, y) such that $l' \leq x < y \leq r'$, the relative order of Touko’s performance matches the ideal: $b_x < b_y$ if and only if $a_x < a_y$.
- A performance b is **k -similar** if it holds its shape for all scripts from 1 to k (i.e., for all intervals $[l_i, r_i]$ where $1 \leq i \leq k$).

Yuu realizes that there are many possible performances b that could satisfy Touko’s strict requirements. To simplify her notes, Yuu decides to encode the set of all k -similar performances using Directed Acyclic Graphs (DAG). She defines a **k -encoding** as a DAG G_k on the set of vertices $1, 2, \dots, n$ such that a performance b satisfies G_k if and only if b is k -similar.

A performance b **satisfies** a DAG G' if for every directed edge $u \rightarrow v$ in G' , we have $b_u < b_v$.

Since Yuu is the only one who can truly understand Touko, she wants to figure out the minimum number of edges among all k -encodings for each k from 1 to q .

Input

The first line contains two integers n and q ($1 \leq n \leq 25000$, $1 \leq q \leq 100000$).

The second line contains n integers $a_1, a_2, \dots, a_n$ which form a permutation of $1, 2, \dots, n$ (Touko’s sister’s ideal performance).

The next q lines each contain two integers l_i and r_i ($1 \leq l_i \leq r_i \leq n$), representing the scripts.

Output

Print q lines. The k -th line should contain a single integer — the minimum number of edges in an optimal k -encoding.

Examples

Example 1

Input:

```
4 3
2 4 1 3
1 3
2 4
1 4
```

Output:

```
2
4
3
```

Note

In the first test case:

- For $k = 1$, the performance must mirror the interval $[1, 3]$. The ideal order is $a_1 = 2, a_2 = 4, a_3 = 1$, meaning $a_3 < a_1 < a_2$. The optimal encoding G_1 has edges $3 \rightarrow 1$ and $1 \rightarrow 2$. Total 2 edges.
- For $k = 2$, Touko must mirror $[1, 3]$ and $[2, 4]$.
- For $k = 3$, adding $[1, 4]$ might actually make some previous edges redundant (transitive reduction).